

EURSKEM AI · ENGINEERING DOCUMENTATION

VOLUME 3

Node Type Catalog

Every registered workflow node type: its category, execution kind, configuration surface, and contract.

Platform: Agentic Workflow Platform

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Typed workflows compiled to LangGraph · zero-token preflight · strict evidence lifecycle

1. Overview

The platform registers **57 node types** in a single registry (`app/nodes/registry.py`). Every type declares three Pydantic schemas — `input_schema` , `output_schema` , `config_schema` — plus an async `run()` ; that is the entire contract. The Builder reads this catalog over `GET /api/node-types` , the runtime compiler uses it to instantiate nodes, and preflight validates templates against each type's declared outputs. Bold fields in the table are required.

Execution kinds: ai: **12** · deterministic: **21** · external: **11** · human: **1** · input: **6** · output: **6**

2. Master Catalog

Node type	Kind	Purpose	Config fields
<code>Echo</code> Control & Flow	deterministic	Renders a template string.	template
<code>Literal</code> Control & Flow	input	Emits a literal config value as its output.	value
<code>SubprocessAgent</code> Control & Flow	deterministic	Launches another saved workflow as a fully independent run, with this step's data becoming that workflow's own inputs, and waits for it to finish before continuing.	workflow , inputs , result_from , result_node , timeout_seconds
<code>TextAssemblerAgent</code> Control & Flow	deterministic	Combines the outputs of several upstream branches into one result once all of them have run — the business 'Join' step after a Parallel Split or Multi-Route.	parts , separator
<code>TransformAgent</code> Control & Flow	ai AI	Pure LLM transform: summarize, classify, rewrite, extract.	mode , model , prompt_template , system_prompt , output_schema , input_fields , instructions , output_fields , language , temperature , max_tokens , max_retries
<code>WorkflowFileLoader</code> Control & Flow	input	Load uploaded workflow files.	files , max_chars_per_file , fail_on_unreadable

		Extracts text from supported PDFs, DOCX, PPTX, Markdown, spreadsheets, and code; images remain stable object-storage references for multimodal nodes.	
AITaskAgent Core Building Blocks		ai AI	Deprecated. TransformAgent's new-style editor now covers the same ground (structured input fields, instructions, typed output schema) plus a <code>`fail_on_error`</code> escape hatch — use that for any new step. Sends content to a language model with a task, an instruction and a typed output schema, and returns a validated structured result. task, instruction, input, context, model, output_fields, language, examples, include_confidence, temperature, max_tokens, max_retries
DataTransformAgent Core Building Blocks		deterministic	Deprecated. TransformAgent now covers the same deterministic operations under mode: deterministic, converging with its AI mode on one config/output schema — use that for any new step. Builds a new object from upstream values using deterministic operations — copy, rename, format, join, coalesce, unit conversion. operations, omit_empty
DecisionAgent Core Building Blocks		deterministic	Evaluates business rules against upstream values and writes named conclusions such as rules, defaults, declared_fields

		human_review, urgency or clarification_required.	
EmailAgent Core Building Blocks	external	Performs one email operation against a configured mailbox connection. Search and read change nothing; draft, reply and send act outside the platform.	connection , operation, query, from_address, subject_contains, unread_only, has_attachments, folder, newer_than_days, max_results, message_id, thread_id
EndAgent Core Building Blocks	output	Collects earlier outputs and presents them as the workflow's final result.	mode, outputs, title, message, chat_message, outcome, route_to, route_to_label, sources, handoff
ExternalActionAgent Core Building Blocks	external	Sends one HTTP request to a URL outside the platform — a REST API call or a webhook notification — and returns the response.	action_type , safety_class , method, url , headers, body, timeout_seconds, allow_unattended_write, approved_by
HumanInLoopAgent Core Building Blocks	human	Pauses the run and waits for a person to approve, edit or reject. The reviewer sees exactly the labelled panels you configure.	question , context_fields, review_panels, review_purpose, editable_content_field, allow_document_override, max_edit_chars, allowed_actions
IntegrationAgent Core Building Blocks	external	Browses, searches, and downloads files from a configured cloud storage connection. Every operation is read-only.	provider , connection , operation, folder_id, file_id, query, page_size, page_token
MCPToolAgent Core Building Blocks	external	Invokes a named tool on a configured MCP server and returns its result as typed workflow data.	server_id , tool , arguments, fail_on_error, timeout_seconds, allow_unattended_write, approved_by, max_read_retries
RouterAgent Core Building Blocks	deterministic	Chooses one outgoing branch (or, in Multi-Route selection, every branch that applies). In field mode it maps a value to a branch;	mode, selection, route_field, branches, cases, fallback, rules, model, prompt, context

			in conditions mode each matching rule group's branch is taken.	
StartAgent Core Building Blocks	input	Declares how information enters the workflow — a form, or a chat message.	mode, title, description, fields, file_fields, sample, chatbot_name, welcome_message, message_placeholder, allow_attachments, suggested_questions	
WorkflowInputAgent Core Building Blocks	input	Declares what enters the workflow and normalises it into a typed object every downstream node can address.	source, fields, sample	
DOCXProposalRenderer Document Rendering & Export	output	Render proposal sections to a styled .docx (corporate, professional, warm).	sections , template, proposal_title , client_name	
ExcelTableExtractor Document Rendering & Export	input	Extract tables from an .xlsx in object storage.	minio_key	
HorizonDOCXProposalRenderer Document Rendering & Export	output	Convert proposal HTML or Markdown into an editable, citation-aware Horizon Europe Part B DOCX with native headings, TOC, tables, figures, page fields, page-limit estimate, and evidence annex.	content , content_format, metadata, citation_registry, evidence_qa, evidence_blockers, include_toc, include_bibliography, include_evidence_annex, page_limit, enforce_page_limit, enable_footnotes	
HorizonHTMLProposalRenderer Document Rendering & Export	output	Convert proposal HTML or Markdown into a citation-aware Horizon Europe Part B PDF with cover, TOC, page checks, and evidence annex.	content , content_format, metadata, citation_registry, evidence_qa, evidence_blockers, include_toc, include_bibliography, include_evidence_annex, page_limit, enforce_page_limit, max_content_characters	
PDFProposalRenderer Document Rendering & Export	output	Render proposal sections to a styled	sections , template, proposal_title , client_name	

		PDF (corporate, professional, warm).	
PDFTextExtractor Document Rendering & Export	input	Extract text from a .pdf in object storage.	minio_key
PowerPointProposalSlides Document Rendering & Export	output	Build a .pptx deck from proposal sections.	sections, proposal_title, client_name
CitationRegistryBuilder Evidence & Retrieval	deterministic	Deterministically reshape acquired full-text documents into a numbered, renderer-ready citation registry with canonical URLs folded into each formatted citation.	documents, require_full_text
ClaimEvidenceVerifier Evidence & Retrieval	ai AI	Verify each linked claim against an exact passage in an immutable source version.	model, minimum_support_confidence, max_source_characters
InternalProjectEvidenceRetrieverAgent Evidence & Retrieval	ai AI	Retrieve partner, pilot, work-plan, budget and approved internal database facts; require an exact source passage and explicit human approval before drafting.	source_registry, source_text, research_briefs, model, max_queries, max_records, max_source_chars, query_internal_index, require_internal_index, internal_index_filters, top_k_candidates, top_n_final
KnowledgeRetrieval Evidence & Retrieval	deterministic	Retrieve secured knowledge through a saved Retrieval Profile without generating an answer.	collection_id, retrieval_profile_id, query, runtime_filters
MinIOEvidenceIngestion Evidence & Retrieval	external	Index acquired full-text sources (from MinIO pages.json) into Weaviate, stamping each chunk with its global citation display_number so retrieved passages carry their footnote	citation_registry, documents, chunk_chars, chunk_overlap_chars, max_chunks_per_source, max_total_chunks, embed_batch_size

			number to the drafter.
PaperQAEvidenceSynthesizerAgent Evidence & Retrieval	deterministic	Run PaperQA2 over already-acquired full-text documents, per claim, for per-document coverage and gap-aware literature synthesis. Never fetches sources itself (no aadd_url) and never certifies evidence.	documents , llm_model , summary_llm_model , embedding_model , evidence_k , max_claims , max_documents_per_claim
PriorProjectRetrieverAgent Evidence & Retrieval	deterministic	Search official CORDIS, LIFE and EIP-AGRI project records for precedents and synergies. Outputs candidates only; exact-passage verification remains mandatory.	research_briefs , sources , provider , max_briefs , max_results_per_source , max_candidates_per_claim , max_total_searches , max_parallel_searches , only_prior_project_track
ProposalTruthGraphAgent Evidence & Retrieval	deterministic	Freeze a drafting-safe truth graph containing only verified or qualified evidence links, plus explicit gaps and approval items.	verified_claims , evidence_gaps , blocking_issues , research_manifest , structured_data_records , internal_evidence_records , evidence_approval_decision
RAGAgent Evidence & Retrieval	ai AI	Hybrid retrieval + grounded answer with citations.	rag_agent_id , model , query , runtime_context , runtime_filters , filters , top_k_candidates , top_n_final , alpha , rerank , compress , generation_prompt
StructuredDatasetRetrieverAgent Evidence & Retrieval	deterministic	Retrieve bounded Eurostat/official structured data with explicit filters, immutable snapshots, row hashes, count reconciliation, and auditable provenance. Records require human review before drafting.	queries , research_briefs , candidate_context , auto_plan_queries , model , max_queries , max_api_calls , max_records , max_response_bytes , request_timeout_seconds , fail_when_no_records

MCPAgent Integrations	external AI	LLM-driven agent loop using MCP tools.	model, objective , system_prompt, max_iterations, temperature, allowed_tools
PythonSnippetAgent Integrations	deterministic	Runs a Python snippet in an isolated sandbox and returns whatever it wrote to `output`.	code , input_fields, output_fields, timeout_seconds, memory_mb, max_output_bytes, fail_on_error
SQLQueryAgent Integrations	external	Runs one read-only SQL SELECT and returns the matched rows.	server_id, sql , params, max_rows, timeout_seconds, fail_on_error
ScientificSkillAgent Integrations	external AI	Scientific synthesis or proposal work guided by approved K-Dense Agent Skills.	model, objective , skills, auto_select, max_skills, system_prompt, temperature, max_tokens
DynamicFigureAgent Multimodal	deterministic	Generate a diagram image for every <code>[[IMAGE PROMPT: ...]]</code> marker and embed it as a data URI the DOCX renderer can render; also emit a caption-only variant for text documents.	content , image_model, size, quality, output_format, max_images, fail_open
FigureEmbedder Multimodal	deterministic	Replace <code>[[IMAGE PROMPT: marker]]</code> placeholders with real embedded images read from object storage, before the proposal is rendered.	content , figures
KimiVisionAgent Multimodal	ai	Analyse an uploaded image with Kimi K3. Image bytes are fetched from object storage and are never written to workflow state.	image , prompt, vision_model, max_completion_tokens
OpenAIImageGenerationAgent Multimodal	ai	Generate an image with an approved OpenAI image model and store the bytes in object storage.	prompt , backend, image_model, size, quality, output_format

CallCoverageMatrixAgent Proposal Engineering	deterministic	Build a deterministic requirement-by-requirement Horizon call coverage matrix and submission gate.	–
ConceptAlternativesAgent Proposal Engineering	ai AI	Generate conservative, balanced, and ambitious Horizon concepts grounded in the approved proposal graph.	model, judge_model, concept_note
ConceptFreezeAgent Proposal Engineering	deterministic	Resolve the human gate's concept decision against the three generated alternatives. Fails closed on an unrecognised id — no LLM call, no default fallback.	alternatives, selected_concept_id
ConsistencyChecker Proposal Engineering	deterministic		block_on_warn
GraphNormalizer Proposal Engineering	deterministic AI		model, max_tokens, concept_note, call_facts
HorizonEvaluationAgent Proposal Engineering	ai AI	Score Excellence, Impact, and Implementation with independent cross-provider evaluators plus deterministic evidence gates.	proposal_text, generator_model, evaluator_models, criterion_threshold, total_threshold
MethodologyEngineeringAgent Proposal Engineering	ai AI	Produce one skill-guided Method Card per frozen-concept objective: method, baseline, validation, uncertainty handling, and failure condition, grounded only in verified claims and known research questions.	model, research_briefs, selected_concept, max_objectives, max_skills_per_card
ProposalEvidenceFactoryAgent Proposal Engineering	ai AI	Verify proposal claims against immutable full-text pages, build an	candidates, documents, search_audit, rejected_candidates, policy,

			auditable citation registry, and fail closed on evidence gaps.	model, citation_style, max_passages_per_document
ProposalSubmissionGate Proposal Engineering	deterministic		Deterministically decide whether a proposal is ready for final human approval or autonomous document export.	proposal_text , evidence_blockers, consistency_gate, consistency_findings, evaluation_threshold_passed, evaluation_total_score, evaluation_blockers, required_headings, minimum_proposal_characters, require_evaluation_pass, block_on_input_needed
BoundedDeepResearchAgent Research & Discovery	deterministic		Run multiple K-Dense-guided bounded research dossiers using a web-search tool-calling loop, with hard job, concurrency, and tool-call limits.	research_briefs , max_jobs, max_parallel_jobs, max_total_tool_calls, max_tool_calls_per_job, max_citations_per_brief, max_candidates_per_claim, max_duration_seconds, max_iterations, max_cost_per_call_usd
ResearchSourceAcquirer Research & Discovery	external		Resolve and store bounded Deep Research citations as immutable HTML or PDF source versions for exact-passage verification.	candidates , policy, max_concurrent_requests, max_sources_per_claim, max_total_sources, request_timeout_seconds, max_redirects, fail_when_none_acquired
ScholarlyCandidateDiscoveryAgent Research & Discovery	external AI		Find scholarly candidate records with multi-query and contradiction searches. Candidates are never treated as verified evidence.	mcp_server, tool, sources, max_results_per_source, max_candidates_per_claim, max_claims, claim_types, require_contradiction_search, model, query_planning_skill, max_skill_prompt_chars, max_duration_seconds
ScientificResearchPlannerAgent Research & Discovery	ai AI		Turn the call, selected concept, and proposal graph into several bounded research briefs, each routed through approved K-Dense skills.	call_context, concept_context, model, max_briefs, max_total_tool_calls, standard_tool_calls, critical_tool_calls, standard_research_model, critical_research_model, max_skills_per_brief
WebSearchAgent Research & Discovery	external		Search the live public web with Auto, Tavily, OpenAI,	query , provider, top_k, fallback_to_openai

or Kimi K3. Results
are candidates, not
verified evidence.

3. Per-Type Reference

Echo — Control & Flow

Renders a template string.

Why it exists: —

Receives: —

Produces: text

Uses AI: No Acts outside the platform: No

Field	Type	Req	Description
template	string	required	

Literal — Control & Flow

Emits a literal config value as its output.

Why it exists: —

Receives: —

Produces: value

Uses AI: No Acts outside the platform: No

Field	Type	Req	Description
value	any	required	

SubprocessAgent — Control & Flow

Launches another saved workflow as a fully independent run, with this step's data becoming that workflow's own inputs, and waits for it to finish before continuing.

Why it exists: A business process that is itself a reusable unit — quoting, onboarding, an approval chain — should be authored once and called from anywhere, the same way a business would delegate it to another team rather than re-explain it every time.

Receives: —

Produces: status, result, child_run_id, child_workflow

Uses AI: No Acts outside the platform: No

Field	Type	Req	Description
workflow	string	required	Which saved workflow to run as a subprocess.
inputs	object	optional	
result_from	string	optional	workflow_output: the child's own declared output: contract. node: one specific node's raw output (set result_node). all_outputs: every node's raw outp
result_node	any	optional	Which child node's output to use, when result_from is 'node'.

timeout_seconds number optional

TextAssemblerAgent — Control & Flow

Combines the outputs of several upstream branches into one result once all of them have run — the business 'Join' step after a Parallel Split or Multi-Route.

Why it exists: A join is naturally deterministic: it does not decide anything, it waits for what already happened and lines it up. No model call means no truncation risk on a long combined document.

Receives: —

Produces: text, part_count, all_parts_present

Uses AI: No **Acts outside the platform:** No

Field	Type	Req	Description
parts	array	optional	
separator	string	optional	

TransformAgent — Control & Flow

Pure LLM transform: summarize, classify, rewrite, extract.

Why it exists: —

Receives: —

Produces: raw, parsed, status, error, data, defaulted

Uses AI: Yes **Acts outside the platform:** No

Field	Type	Req	Description
mode	string	optional	
model	string	optional	
prompt_template	any	optional	
system_prompt	any	optional	
output_schema	object	optional	
input_fields	array	optional	
instructions	string	optional	
output_fields	array	optional	
language	any	optional	
temperature	number	optional	
max_tokens	integer	optional	
max_retries	integer	optional	
fail_on_error	boolean	optional	

<code>operations</code>	array	optional
<code>omit_empty</code>	boolean	optional

WorkflowFileLoader — Control & Flow

Load uploaded workflow files. Extracts text from supported PDFs, DOCX, PPTX, Markdown, spreadsheets, and code; images remain stable object-storage references for multimodal nodes.

Why it exists: —

Receives: —

Produces: text, files, image_files, total_files, text_file_count, image_count

Uses AI: No **Acts outside the platform:** No

Field	Type	Req	Description
<code>files</code>	any	required	
<code>max_chars_per_file</code>	integer	optional	
<code>fail_on_unreadable</code>	boolean	optional	

AITaskAgent — Core Building Blocks

Deprecated. TransformAgent's new-style editor now covers the same ground (structured input fields, instructions, typed output schema) plus a `fail_on_error` escape hatch — use that for any new step. Sends content to a language model with a task, an instruction and a typed output schema, and returns a validated structured result.

Why it exists: Replaces per-purpose AI agents. The prompt, labels, schema, language policy and model are configuration, so a new business behaviour needs no new node type.

Receives: —

Produces: result, text, status, error, confidence, reasoning, detected_language, model_used, attempts

Uses AI: Yes **Acts outside the platform:** No

Field	Type	Req	Description
<code>task</code>	string	optional	What kind of AI capability this step performs — extract, classify, summarize, translate, draft a reply, etc.
<code>instruction</code>	string	optional	The task in your own words — what you want the model to do with the input.
<code>input</code>	string	optional	What this step reads — normally a reference to an earlier step's output, written by the mapping picker.
<code>context</code>	object	optional	Extra labelled context blocks (each templated) to keep the main input readable when the task needs several sources.
<code>model</code>	string	optional	Which language model runs this step. 'auto' lets the platform route to the best fit for the task.
<code>output_fields</code>	array	optional	The structured output shape this step must return — required for extract/classify tasks.
<code>language</code>	any	optional	

			Which language the model should reply in, or whether to match the input's language.
<code>examples</code>	array	optional	Optional worked examples shown to the model to steer its output format and tone.
<code>include_confidence</code>	boolean	optional	Adds a confidence score and reasoning to the output, so a Decision/Router step can route uncertain answers to a human.
<code>temperature</code>	number	optional	Higher values make the model's answer more varied; 0 is the most consistent.
<code>max_tokens</code>	integer	optional	Upper bound on how long the model's answer may be.
<code>max_retries</code>	integer	optional	How many times to retry if the model's output fails to validate against the output schema.
<code>fail_on_error</code>	boolean	optional	When off, a model failure becomes a routable status instead of stopping the run — lets a Decision/Router step send it to a human instead.
<code>reasoning_effort</code>	any	optional	Optional reasoning-effort override for models that support it (low/medium/high).

DataTransformAgent — Core Building Blocks

Deprecated. TransformAgent now covers the same deterministic operations under mode: deterministic, converging with its AI mode on one config/output schema — use that for any new step. Builds a new object from upstream values using deterministic operations — copy, rename, format, join, coalesce, unit conversion.

Why it exists: These operations are exact. Using a model for them would add cost, latency and a failure mode for no benefit.

Receives: —

Produces: data, defaulted

Uses AI: No **Acts outside the platform:** No

Field	Type	Req	Description
<code>operations</code>	array	optional	Deterministic operations (copy, rename, format, join, coalesce, unit conversion) that build the output object.
<code>omit_empty</code>	boolean	optional	Drop keys whose computed value is empty/None — useful when building a payload for an API that rejects explicit nulls.

DecisionAgent — Core Building Blocks

Evaluates business rules against upstream values and writes named conclusions such as human_review, urgency or clarification_required.

Why it exists: Business policy should be inspectable and repeatable. Rules cost no tokens, never drift, and record exactly why each conclusion held.

Receives: —

Produces: decisions, matched_rules, explanation, summary

Uses AI: No **Acts outside the platform:** No

Field	Type	Req	Description
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rules	array	optional	Business rules evaluated in order — the first matching rule's conclusions win.
defaults	object	optional	What each conclusion is before any rule runs — guarantees a downstream step always has a value to read, even when no rule fires.
declared_fields	array	optional	Extra output field names this step promises to produce, beyond what defaults/rules already declare.

EmailAgent — Core Building Blocks

Performs one email operation against a configured mailbox connection. Search and read change nothing; draft, reply and send act outside the platform.

Why it exists: One capability with an operation selector, instead of a node type per provider and verb. Which mailbox and which credentials is a deployment decision, not workflow content.

Receives: —

Produces: operation, provider, connection, messages, message, message_count, draft_id, sent_message_id, deduplicated

Uses AI: No **Acts outside the platform:** Yes

Field	Type	Req	Description
connection	string	required	Which configured mailbox connection this step acts against.
operation	string	optional	The email operation to perform: search/read change nothing; create_draft/reply/send take action.
query	string	optional	Search query, in search operations.
from_address	any	optional	Filter search results to this sender.
subject_contains	any	optional	Filter search results to a subject substring.
unread_only	boolean	optional	Only match unread messages, in search operations.
has_attachments	boolean	optional	Only match messages with attachments, in search operations.
folder	string	optional	Mailbox folder to search.
newer_than_days	any	optional	Only match messages newer than this many days.
max_results	integer	optional	Maximum number of search results to return.
message_id	any	optional	Message to read or reply to.
thread_id	any	optional	Thread to reply within.
to	array	optional	Recipients, in create_draft/reply/send operations.
cc	array	optional	CC recipients.
bcc	array	optional	BCC recipients.
subject	string	optional	Message subject, in create_draft/send operations.
body	string	optional	Plain-text message body — normally a template reference.

<code>body_html</code>	any	optional	Optional HTML message body.
<code>allow_unattended_write</code>	boolean	optional	Explicit statement that a send/reply may happen without a human review step in front of it. Ignored for search/read/create_draft.
<code>approved_by</code>	any	optional	Node id of the HumanInLoopAgent review that gates a send/reply. When set, only that node's decision counts.

EndAgent — Core Building Blocks

Collects earlier outputs and presents them as the workflow's final result.

Why it exists: Gives every workflow a clear, visible destination for its result.

Receives: —

Produces: result

Uses AI: No **Acts outside the platform:** No

Field	Type	Req	Description
<code>mode</code>	string	optional	
<code>outputs</code>	array	optional	
<code>title</code>	string	optional	
<code>message</code>	string	optional	
<code>chat_message</code>	string	optional	
<code>outcome</code>	string	optional	
<code>route_to</code>	any	optional	
<code>route_to_label</code>	any	optional	
<code>sources</code>	any	optional	
<code>handoff</code>	object	optional	

ExternalActionAgent — Core Building Blocks

Sends one HTTP request to a URL outside the platform — a REST API call or a webhook notification — and returns the response.

Why it exists: Not every outward action goes through a connected MCP server. This is the generic escape hatch for a specific URL, kept separate so it never gets confused with the classified, discoverable tools MCP already provides.

Receives: —

Produces: status, safety_class, response_status, response_body, duration_s, deduplicated, error, error_code, retryable

Uses AI: No **Acts outside the platform:** Yes

Field	Type	Req	Description
<code>action_type</code>	string	required	rest_api: a request/response call whose response matters downstream. webhook: a fire-and-forget outbound notification.

<code>safety_class</code>	string	required	How this call is classified: read (nothing changes), write (a business record changes), external_action (a generic outward effect — a notification, a
<code>method</code>	string	optional	HTTP method. Webhooks default to POST; a REST API call may use any of these.
<code>url</code>	string	required	Target URL — template-resolved like any other config string.
<code>headers</code>	object	optional	
<code>body</code>	any	optional	Request body, sent as JSON. Usually a mapped object from an earlier step.
<code>timeout_seconds</code>	number	optional	
<code>allow_unattended_write</code>	boolean	optional	Explicit statement that this call may run without a human review step in front of it, when <code>safety_class</code> is write or external_action.
<code>approved_by</code>	any	optional	Node id of the HumanInLoopAgent review that gates this call. When set, only that node's decision counts.

HumanInLoopAgent — Core Building Blocks

Pauses the run and waits for a person to approve, edit or reject. The reviewer sees exactly the labelled panels you configure.

Why it exists: The author decides which actions may happen automatically. A gate in front of an external action is how that boundary becomes visible on the canvas instead of living inside a prompt.

Receives: —

Produces: decision, reason, content, edited_content, content_overridden

Uses AI: No **Acts outside the platform:** No

Field	Type	Req	Description
<code>question</code>	string	required	The question shown to the reviewer, templated with upstream values.
<code>context_fields</code>	array	optional	Raw upstream fields to show the reviewer as fallback context, when <code>review_panels</code> isn't set.
<code>review_panels</code>	array	optional	Labelled panels the reviewer sees, in place of the raw <code>context_fields</code> list.
<code>review_purpose</code>	string	optional	Business-language explanation of why this gate exists, shown above the decision buttons.
<code>editable_content_field</code>	any	optional	Optional state path shown in the rich-text editor. When the human saves an edit, the value at this path is replaced before downstream nodes run. If om
<code>allow_document_override</code>	boolean	optional	Allow one uploaded, text-extractable document to replace the editor content before the human continues.
<code>max_edit_chars</code>	integer	optional	Maximum plain-text characters accepted from the editor.
<code>allowed_actions</code>	array	optional	

IntegrationAgent — Core Building Blocks

Browses, searches, and downloads files from a configured cloud storage connection. Every operation is read-only.

Why it exists: One capability with a provider and operation selector, instead of a node type per cloud-storage vendor. Which account and which credentials is a deployment decision, not workflow content.

Receives: —

Produces: operation, provider, connection, status, files, file, first, count, found, downloaded_file, downloaded_files, next_page_token, error, error_code, retryable

Uses AI: No **Acts outside the platform:** Yes

Field	Type	Req	Description
provider	string	required	Which cloud-storage provider this step reads from.
connection	string	required	Which configured account connection this step acts against.
operation	string	optional	The file operation to perform: list_folder/search_files browse; select_file/select_folder/get_file target one or more items.
folder_id	any	optional	Folder(s) to list, to scope a search to, or to select — root when empty for list_folder/search_files.
file_id	any	optional	File(s) to select or download.
query	string	optional	Search text, in search_files.
page_size	integer	optional	
page_token	any	optional	Opaque pagination cursor from a previous page.

MCPToolAgent — Core Building Blocks

Invokes a named tool on a configured MCP server and returns its result as typed workflow data.

Why it exists: Business systems are reached through MCP, not through node types. A new CRM capability is a new MCP tool; the Builder discovers it with no change here and no change in the frontend.

Receives: —

Produces: server, tool, operation, status, data, first, text, count, found, is_structured, mode, duration_s, deduplicated, error, error_code, retryable, suggested_action

Uses AI: No **Acts outside the platform:** Yes

Field	Type	Req	Description
server_id	string	required	Which configured MCP server connection this step calls.
tool	string	required	Which tool on that server to invoke.
arguments	object	optional	Arguments for the tool, matching its discovered input schema — normally template references from the field picker.
fail_on_error	boolean	optional	When off, a tool failure becomes a routable status instead of stopping the run.
timeout_seconds	any	optional	Upper bound on this one call, overriding the connection's default timeout.

<code>allow_unattended_write</code>	boolean	optional	Explicit statement that this write may happen without a human review step in front of it. Does not override the connection's own write policy.
<code>approved_by</code>	any	optional	Node id of the HumanInLoopAgent review that gates this write. When set, only that node's decision counts.
<code>max_read_retries</code>	integer	optional	Retries for read-only tool calls. Writes are never retried here.

RouterAgent — Core Building Blocks

Chooses one outgoing branch (or, in Multi-Route selection, every branch that applies). In field mode it maps a value to a branch; in conditions mode each matching rule group's branch is taken.

Why it exists: Routing is where a business process becomes visible. Branch labels appear on the canvas edges, and the chosen branch carries its reason.

Receives: —

Produces: route, routes, reason, route_value, explanation, matched_conditions, used_fallback

Uses AI: No **Acts outside the platform:** No

Field	Type	Req	Description
<code>mode</code>	string	optional	How this step decides a branch: field (map a value to a branch), conditions (first matching rule group), rule (legacy expressions), or llm (ask a mode
<code>selection</code>	string	optional	How many branches this step can take at once: single (choose one, the default) or multi (choose one or more — Multi-Route).
<code>route_field</code>	any	optional	The value to branch on, in field mode — e.g. an upstream step's classified intent.
<code>branches</code>	object	optional	value → route name, in field mode. Several values may share one route name.
<code>cases</code>	array	optional	Ordered rule groups, in conditions mode — the first matching group's route wins.
<code>fallback</code>	any	optional	Route used when nothing else matches. Leaving this unset can fail a run on an unexpected value — preflight flags a missing fallback.
<code>rules</code>	array	optional	Legacy string-expression rules, in rule mode.
<code>model</code>	any	optional	Which model chooses the route, in llm mode.
<code>prompt</code>	any	optional	What to ask the model when picking a route, in llm mode.
<code>context</code>	any	optional	Templated context passed to the model, in llm mode.

StartAgent — Core Building Blocks

Declares how information enters the workflow — a form, or a chat message.

Why it exists: Gives the workflow a visible, understandable starting point.

Receives: —

Produces: data, message, attachments, missing

Uses AI: No **Acts outside the platform:** No

Field	Type	Req	Description
mode	string	optional	
title	string	optional	
description	string	optional	
fields	array	optional	
file_fields	array	optional	
sample	object	optional	
chatbot_name	string	optional	
welcome_message	string	optional	
message_placeholder	string	optional	
allow_attachments	boolean	optional	
suggested_questions	array	optional	

WorkflowInputAgent — Core Building Blocks

Declares what enters the workflow and normalises it into a typed object every downstream node can address.

Why it exists: Gives the process a visible starting point and makes the mapping picker and rule editor typed from the first node onward.

Receives: —

Produces: data, source, missing

Uses AI: No **Acts outside the platform:** No

Field	Type	Req	Description
source	string	optional	Where this workflow's input data comes from.
fields	array	optional	The declared shape of what enters the workflow — every downstream step addresses these by name.
sample	object	optional	Sample payload used by the Builder's Test tab and Simulator to exercise the workflow before any real integration is connected.

DOCXProposalRenderer — Document Rendering & Export

Render proposal sections to a styled .docx (corporate, professional, warm).

Why it exists: —

Receives: —

Produces: minio_key, byte_size, template_used

Uses AI: No **Acts outside the platform:** No

Field	Type	Req	Description
sections	object	required	

Map of section_name -> section_text (Markdown). Templated from upstream nodes.

template string optional

proposal_title string **required**

client_name string **required**

ExcelTableExtractor — Document Rendering & Export

Extract tables from an .xlsx in object storage.

Why it exists: —

Receives: —

Produces: tables, sheet_count, total_rows

Uses AI: No Acts outside the platform: No

Field	Type	Req	Description
minio_key	string	required	

HorizonDOCXProposalRenderer — Document Rendering & Export

Convert proposal HTML or Markdown into an editable, citation-aware Horizon Europe Part B DOCX with native headings, TOC, tables, figures, page fields, page-limit estimate, and evidence annex.

Why it exists: —

Receives: —

Produces: minio_key, docx_key, source_html_key, byte_size, source_html_byte_size, estimated_page_count, page_count, page_count_basis, page_limit, docx_sha256, source_html_sha256, template_used, template_version, table_of_contents, warnings, embedded_image_count, image_placeholder_count, submission_ready

Uses AI: No Acts outside the platform: No

Field	Type	Req	Description
content	string	required	
content_format	string	optional	
metadata	any	optional	
citation_registry	any	optional	
evidence_qa	any	optional	
evidence_blockers	any	optional	
include_toc	boolean	optional	
include_bibliography	boolean	optional	
include_evidence_annex	boolean	optional	
page_limit	any	optional	

<code>enforce_page_limit</code>	boolean	optional
<code>enable_footnotes</code>	boolean	optional
<code>max_content_characters</code>	integer	optional
<code>max_embedded_image_bytes</code>	integer	optional

HorizonHTMLProposalRenderer — Document Rendering & Export

Convert proposal HTML or Markdown into a citation-aware Horizon Europe Part B PDF with cover, TOC, page checks, and evidence annex.

Why it exists: —

Receives: —

Produces: minio_key, pdf_key, html_key, byte_size, html_byte_size, page_count, page_limit, html_sha256, pdf_sha256, template_used, template_version, table_of_contents, warnings, submission_ready

Uses AI: No **Acts outside the platform:** No

Field	Type	Req	Description
<code>content</code>	string	required	
<code>content_format</code>	string	optional	
<code>metadata</code>	any	optional	
<code>citation_registry</code>	any	optional	
<code>evidence_qa</code>	any	optional	
<code>evidence_blockers</code>	any	optional	
<code>include_toc</code>	boolean	optional	
<code>include_bibliography</code>	boolean	optional	
<code>include_evidence_annex</code>	boolean	optional	
<code>page_limit</code>	any	optional	
<code>enforce_page_limit</code>	boolean	optional	
<code>max_content_characters</code>	integer	optional	

PDFProposalRenderer — Document Rendering & Export

Render proposal sections to a styled PDF (corporate, professional, warm).

Why it exists: —

Receives: —

Produces: minio_key, byte_size, template_used

Uses AI: No **Acts outside the platform:** No

Field	Type	Req	Description
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sections	object	required	Map of section_name -> section_text. Templated from upstream nodes.
template	string	optional	
proposal_title	string	required	
client_name	string	required	

PDFTextExtractor — Document Rendering & Export

Extract text from a .pdf in object storage.

Why it exists: —

Receives: —

Produces: pages, page_count

Uses AI: No Acts outside the platform: No

Field	Type	Req	Description
minio_key	string	required	

PowerPointProposalSlides — Document Rendering & Export

Build a .pptx deck from proposal sections.

Why it exists: —

Receives: —

Produces: minio_key, slide_count, byte_size

Uses AI: No Acts outside the platform: No

Field	Type	Req	Description
sections	object	required	
proposal_title	string	required	
client_name	string	required	

CitationRegistryBuilder — Evidence & Retrieval

Deterministically reshape acquired full-text documents into a numbered, renderer-ready citation registry with canonical URLs folded into each formatted citation.

Why it exists: —

Receives: —

Produces: citation_registry, claim_to_numbers, drafting_guide, total_sources, total_documents_considered, excluded_documents

Uses AI: No Acts outside the platform: No

Field	Type	Req	Description
documents	any	optional	
require_full_text	boolean	optional	

ClaimEvidenceVerifier — Evidence & Retrieval

Verify each linked claim against an exact passage in an immutable source version.

Why it exists: —

Receives: —

Produces: claims_checked, relations_created, supported_claims, contradicted_claims, unverified_claims, findings

Uses AI: Yes **Acts outside the platform:** No

Field	Type	Req	Description
model	string	optional	
minimum_support_confidence	number	optional	
max_source_characters	integer	optional	

InternalProjectEvidenceRetrieverAgent — Evidence & Retrieval

Retrieve partner, pilot, work-plan, budget and approved internal database facts; require an exact source passage and explicit human approval before drafting.

Why it exists: —

Receives: —

Produces: records, approved_records, pending_human_approval, rejected_facts, unresolved_questions, internal_index_used, verification_status, report

Uses AI: Yes **Acts outside the platform:** No

Field	Type	Req	Description
source_registry	any	required	
source_text	string	required	
research_briefs	any	optional	
model	string	optional	
max_queries	integer	optional	
max_records	integer	optional	
max_source_chars	integer	optional	
query_internal_index	boolean	optional	
require_internal_index	boolean	optional	
internal_index_filters	object	optional	
top_k_candidates	integer	optional	
top_n_final	integer	optional	

KnowledgeRetrieval — Evidence & Retrieval

Retrieve secured knowledge through a saved Retrieval Profile without generating an answer.

Why it exists: —

Receives: —

Produces: retrieved_chunks, citations, context, retrieval_trace_id, collection_id, resolved_index_id, retrieval_profile_id, candidate_count, context_count, timings_ms, resolved_resources

Uses AI: No **Acts outside the platform:** No

Field	Type	Req	Description
collection_id	string	required	
retrieval_profile_id	string	required	
query	string	required	
runtime_filters	object	optional	

MinIOEvidenceIngestion — Evidence & Retrieval

Index acquired full-text sources (from MinIO pages.json) into Weaviate, stamping each chunk with its global citation display_number so retrieved passages carry their footnote number to the drafter.

Why it exists: —

Receives: —

Produces: sources_indexed, sources_skipped, chunks_written, collection_id, session_id, skipped_detail

Uses AI: No **Acts outside the platform:** Yes

Field	Type	Req	Description
citation_registry	any	optional	
documents	any	optional	
chunk_chars	integer	optional	
chunk_overlap_chars	integer	optional	
max_chunks_per_source	integer	optional	
max_total_chunks	integer	optional	
embed_batch_size	integer	optional	

PaperQAEvidenceSynthesizerAgent — Evidence & Retrieval

Run PaperQA2 over already-acquired full-text documents, per claim, for per-document coverage and gap-aware literature synthesis. Never fetches sources itself (no aadd_url) and never certifies evidence.

Why it exists: —

Receives: —

Produces: results, claims_processed, claims_skipped_no_claim_text, total_cost_usd, verification_status, note

Uses AI: No **Acts outside the platform:** No

Field	Type	Req	Description
documents	any	required	

llm_model	string	optional
summary_llm_model	string	optional
embedding_model	string	optional
evidence_k	integer	optional
max_claims	integer	optional
max_documents_per_claim	integer	optional

PriorProjectRetrieverAgent — Evidence & Retrieval

Search official CORDIS, LIFE and EIP-AGRI project records for precedents and synergies. Outputs candidates only; exact-passage verification remains mandatory.

Why it exists: —

Receives: —

Produces: candidates, search_audit, projects_found, searches_completed, searches_failed, verification_status, note, report

Uses AI: No **Acts outside the platform:** No

Field	Type	Req	Description
research_briefs	any	required	
sources	array	optional	
provider	string	optional	
max_briefs	integer	optional	
max_results_per_source	integer	optional	
max_candidates_per_claim	integer	optional	
max_total_searches	integer	optional	
max_parallel_searches	integer	optional	
only_prior_project_track	boolean	optional	

ProposalTruthGraphAgent — Evidence & Retrieval

Freeze a drafting-safe truth graph containing only verified or qualified evidence links, plus explicit gaps and approval items.

Why it exists: —

Receives: —

Produces: truth_graph, integrity_sha256, verified_claim_ids, qualified_claim_ids, excluded_claim_ids, human_review_queue, evidence_gaps, blocking_issues, drafting_allowed, approval_required, report

Uses AI: No **Acts outside the platform:** No

Field	Type	Req	Description
verified_claims	any	required	

evidence_gaps	any	optional
blocking_issues	any	optional
research_manifest	any	optional
structured_data_records	any	optional
internal_evidence_records	any	optional
evidence_approval_decision	string	optional

RAGAgent — Evidence & Retrieval

Hybrid retrieval + grounded answer with citations.

Why it exists: —

Receives: —

Produces: query, answer, citations, sources, relevant_context, answering_model, resolved_answering_model, retrievals, rewritten_query, grounding_for_drafter, retrieval_trace_id, collection_id, resolved_index_id, rag_agent_id, retrieval_profile_id, generation_profile_id, candidate_count, context_count, timings_ms, resolved_resources, generation

Uses AI: Yes **Acts outside the platform:** No

Field	Type	Req	Description
rag_agent_id	any	optional	Saved RAG Agent resource. New workflows should set this.
model	any	optional	
query	string	required	
runtime_context	any	optional	Additional runtime info to accompany the query (e.g. a customer record from an upstream node). Not indexed, not used for retrieval filtering — kept di
runtime_filters	object	optional	Additional typed metadata filters; security scope is never overridable.
filters	object	optional	
top_k_candidates	integer	optional	
top_n_final	integer	optional	
alpha	number	optional	
rerank	boolean	optional	
compress	boolean	optional	
generation_prompt	string	optional	

StructuredDatasetRetrieverAgent — Evidence & Retrieval

Retrieve bounded Eurostat/official structured data with explicit filters, immutable snapshots, row hashes, count reconciliation, and auditable provenance. Records require human review before drafting.

Why it exists: —

Receives: —

Produces: retrieval_contracts, records, quantitative_evidence_registry, audit, unresolved_questions, records_retrieved, queries_completed, queries_failed, verification_status, report

Uses AI: No **Acts outside the platform:** No

Field	Type	Req	Description
queries	any	optional	
research_briefs	any	optional	
candidate_context	any	optional	
auto_plan_queries	boolean	optional	
model	string	optional	
max_queries	integer	optional	
max_api_calls	integer	optional	
max_records	integer	optional	
max_response_bytes	integer	optional	
request_timeout_seconds	number	optional	
fail_when_no_records	boolean	optional	

MCPAgent — Integrations

LLM-driven agent loop using MCP tools.

Why it exists: —

Receives: —

Produces: answer, tool_calls, iterations_used, completed

Uses AI: Yes **Acts outside the platform:** Yes

Field	Type	Req	Description
model	string	optional	
objective	string	required	
system_prompt	string	optional	
max_iterations	integer	optional	
temperature	number	optional	
allowed_tools	any	optional	

PythonSnippetAgent — Integrations

Runs a Python snippet in an isolated sandbox and returns whatever it wrote to `output`.

Why it exists: Some steps are just a few lines of real code — a calculation, a reshape a Transform's operations don't cover — and forcing that through an LLM call would be slower, costlier, and less predictable than just running it.

Receives: —

Produces: status, result, stdout, stderr, error, duration_s

Uses AI: No **Acts outside the platform:** No

Field	Type	Req	Description
code	string	required	Python source. Reads inputs from the `inputs` dict; writes results into the `output` dict — both are already bound, no import needed.
input_fields	object	optional	
output_fields	array	optional	
timeout_seconds	number	optional	
memory_mb	integer	optional	
max_output_bytes	integer	optional	
fail_on_error	boolean	optional	When off, a snippet failure becomes a routable status instead of stopping the run.

SQLQueryAgent — Integrations

Runs one read-only SQL SELECT and returns the matched rows.

Why it exists: The classified lookups (Customer Search, Order Search, ...) cover the common cases; this is the escape hatch for a specific query they don't — still read-only, still bounded, never a way to write.

Receives: —

Produces: status, rows, count, found, first, truncated, error, error_code, retryable

Uses AI: No **Acts outside the platform:** Yes

Field	Type	Req	Description
server_id	string	optional	Which configured MCP connection exposes query_readonly.
sql	string	required	A single SELECT statement. Use %(name)s placeholders for any mapped value — never build the string with the value already inside it, and never map a v
params	object	optional	Named values for the query's %(name)s placeholders.
max_rows	integer	optional	
timeout_seconds	number	optional	
fail_on_error	boolean	optional	When off, a query failure becomes a routable status instead of stopping the run.

ScientificSkillAgent — Integrations

Scientific synthesis or proposal work guided by approved K-Dense Agent Skills.

Why it exists: —

Receives: —

Produces: answer, skills_used, skill_versions, selection_reason

Uses AI: Yes **Acts outside the platform:** Yes

Field	Type	Req	Description
model	string	optional	
objective	string	required	
skills	array	optional	Approved K-Dense skill names, one per line in the Builder. Leave empty to use deterministic automatic selection.
auto_select	boolean	optional	
max_skills	integer	optional	
system_prompt	string	optional	
temperature	number	optional	
max_tokens	integer	optional	

DynamicFigureAgent — Multimodal

Generate a diagram image for every `[[IMAGE PROMPT: ...]]` marker and embed it as a data URI the DOCX renderer can render; also emit a caption-only variant for text documents.

Why it exists: —

Receives: —

Produces: illustrated_content, captioned_content, figures, markers_found, images_generated

Uses AI: No **Acts outside the platform:** No

Field	Type	Req	Description
content	string	required	
image_model	string	optional	
size	string	optional	
quality	string	optional	
output_format	string	optional	
max_images	integer	optional	
fail_open	boolean	optional	

FigureEmbedder — Multimodal

Replace `[[IMAGE PROMPT: marker]]` placeholders with real embedded images read from object storage, before the proposal is rendered.

Why it exists: —

Receives: —

Produces: content, embedded_count, unmatched_figures, missing_images

Uses AI: No **Acts outside the platform:** No

Field	Type	Req	Description
content	string	required	
figures	array	optional	

KimiVisionAgent — Multimodal

Analyse an uploaded image with Kimi K3. Image bytes are fetched from object storage and are never written to workflow state.

Why it exists: —

Receives: —

Produces: analysis, provider, model, image_name, minio_key, content_type, byte_size, input_tokens, output_tokens, skipped

Uses AI: No **Acts outside the platform:** No

Field	Type	Req	Description
image	any	required	
prompt	string	optional	
vision_model	string	optional	Kimi vision model.
max_completion_tokens	integer	optional	

OpenAIImageGenerationAgent — Multimodal

Generate an image with an approved OpenAI image model and store the bytes in object storage.

Why it exists: —

Receives: —

Produces: generated, provider, model, minio_key, content_type, byte_size, revised_prompt

Uses AI: No **Acts outside the platform:** No

Field	Type	Req	Description
prompt	string	required	
backend	string	optional	Disable image creation or use OpenAI image generation.
image_model	string	optional	Approved OpenAI image model for this node.
size	string	optional	
quality	string	optional	
output_format	string	optional	

CallCoverageMatrixAgent — Proposal Engineering

Build a deterministic requirement-by-requirement Horizon call coverage matrix and submission gate.

Why it exists: —

Receives: —

Produces: rows, addressed, partial, missing, coverage_percent, blocking_requirement_ids, submission_blocked

Uses AI: No **Acts outside the platform:** No

No configurable fields.

ConceptAlternativesAgent — Proposal Engineering

Generate conservative, balanced, and ambitious Horizon concepts grounded in the approved proposal graph.

Why it exists: —

Receives: —

Produces: alternatives, recommended_concept_id

Uses AI: Yes **Acts outside the platform:** No

Field	Type	Req	Description
model	string	optional	
judge_model	any	optional	
concept_note	string	optional	

ConceptFreezeAgent — Proposal Engineering

Resolve the human gate's concept decision against the three generated alternatives. Fails closed on an unrecognised id — no LLM call, no default fallback.

Why it exists: —

Receives: —

Produces: selected_concept

Uses AI: No **Acts outside the platform:** No

Field	Type	Req	Description
alternatives	any	required	
selected_concept_id	string	required	

ConsistencyChecker — Proposal Engineering

Why it exists: —

Receives: —

Produces: gate, findings, report

Uses AI: No **Acts outside the platform:** No

Field	Type	Req	Description
block_on_warn	boolean	optional	

GraphNormalizer — Proposal Engineering

Why it exists: —

Receives: —

Produces: counts, warnings, report

Uses AI: Yes **Acts outside the platform:** No

Field	Type	Req	Description
model	string	optional	
max_tokens	integer	optional	
concept_note	any	optional	
call_facts	any	optional	

HorizonEvaluationAgent — Proposal Engineering

Score Excellence, Impact, and Implementation with independent cross-provider evaluators plus deterministic evidence gates.

Why it exists: —

Receives: —

Produces: prompt_version, generator_model, evaluator_models, criteria, total_score, threshold_passed, coverage_percent, deterministic_blockers, high_disagreement_criteria

Uses AI: Yes **Acts outside the platform:** No

Field	Type	Req	Description
proposal_text	string	required	
generator_model	any	optional	
evaluator_models	array	optional	
criterion_threshold	number	optional	
total_threshold	number	optional	

MethodologyEngineeringAgent — Proposal Engineering

Produce one skill-guided Method Card per frozen-concept objective: method, baseline, validation, uncertainty handling, and failure condition, grounded only in verified claims and known research questions.

Why it exists: —

Receives: —

Produces: method_cards, objectives_processed, skills_manifest, skill_versions

Uses AI: Yes **Acts outside the platform:** No

Field	Type	Req	Description
model	string	optional	
research_briefs	any	optional	
selected_concept	any	optional	

max_objectives	integer	optional
max_skills_per_card	integer	optional

ProposalEvidenceFactoryAgent — Proposal Engineering

Verify proposal claims against immutable full-text pages, build an auditable citation registry, and fail closed on evidence gaps.

Why it exists: —

Receives: —

Produces: verified_claims, citation_registry, claim_evidence_links, quantitative_evidence_registry, proposal_ready_cited_markdown, bibliography_markdown, bibliography_bibtex, search_audit, rejected_candidates, evidence_gaps, qa_report, blocking_issues, graph_sources_added, graph_relations_added, report

Uses AI: Yes **Acts outside the platform:** No

Field	Type	Req	Description
candidates	any	required	
documents	any	required	
search_audit	any	optional	
rejected_candidates	any	optional	
policy	any	optional	
model	string	optional	
citation_style	string	optional	
max_passages_per_document	integer	optional	

ProposalSubmissionGate — Proposal Engineering

Deterministically decide whether a proposal is ready for final human approval or autonomous document export.

Why it exists: —

Receives: —

Produces: status, submission_ready, blockers, warnings, checks, input_needed_count, proposal_characters, evaluation_total_score, report

Uses AI: No **Acts outside the platform:** No

Field	Type	Req	Description
proposal_text	string	required	
evidence_blockers	any	optional	
consistency_gate	string	optional	
consistency_findings	any	optional	
evaluation_threshold_passed	any	optional	

evaluation_total_score	any	optional
evaluation_blockers	any	optional
required_headings	array	optional
minimum_proposal_characters	integer	optional
require_evaluation_pass	boolean	optional
block_on_input_needed	boolean	optional

BoundedDeepResearchAgent — Research & Discovery

Run multiple K-Dense-guided bounded research dossiers using a web-search tool-calling loop, with hard job, concurrency, and tool-call limits.

Why it exists: —

Receives: —

Produces: dossiers, candidates, search_audit, jobs_completed, jobs_failed, failures, total_tool_call_budget, actual_tool_calls, input_tokens, output_tokens, research_manifest

Uses AI: No **Acts outside the platform:** No

Field	Type	Req	Description
research_briefs	any	required	
max_jobs	integer	optional	
max_parallel_jobs	integer	optional	
max_total_tool_calls	integer	optional	
max_tool_calls_per_job	integer	optional	
max_citations_per_brief	integer	optional	
max_candidates_per_claim	integer	optional	
max_duration_seconds	number	optional	
max_iterations	integer	optional	
max_cost_per_call_usd	number	optional	

ResearchSourceAcquirer — Research & Discovery

Resolve and store bounded Deep Research citations as immutable HTML or PDF source versions for exact-passage verification.

Why it exists: —

Receives: —

Produces: candidates_processed, full_text_documents_acquired, documents, rejected_candidates, report

Uses AI: No **Acts outside the platform:** Yes

Field	Type	Req	Description
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candidates	any	required
policy	any	optional
max_concurrent_requests	integer	optional
max_sources_per_claim	integer	optional
max_total_sources	integer	optional
request_timeout_seconds	number	optional
max_redirects	integer	optional
fail_when_none_acquired	boolean	optional

ScholarlyCandidateDiscoveryAgent — Research & Discovery

Find scholarly candidate records with multi-query and contradiction searches. Candidates are never treated as verified evidence.

Why it exists: —

Receives: —

Produces: candidates_found, claims_searched, candidates, search_audit, report, timed_out, sources_added, claims_linked, sources

Uses AI: Yes **Acts outside the platform:** Yes

Field	Type	Req	Description
mcp_server	string	optional	
tool	string	optional	
sources	array	optional	
max_results_per_source	integer	optional	
max_candidates_per_claim	integer	optional	
max_claims	integer	optional	
claim_types	array	optional	
require_contradiction_search	boolean	optional	
model	any	optional	
query_planning_skill	any	optional	
max_skill_prompt_chars	integer	optional	
max_duration_seconds	number	optional	

ScientificResearchPlannerAgent — Research & Discovery

Turn the call, selected concept, and proposal graph into several bounded research briefs, each routed through approved K-Dense skills.

Why it exists: —

Receives: —

Produces: research_briefs, brief_count, total_tool_call_budget, skills_manifest, skill_versions, unresolved_questions, governance_rules

Uses AI: Yes **Acts outside the platform:** No

Field	Type	Req	Description
call_context	any	optional	
concept_context	any	optional	
model	string	optional	
max_briefs	integer	optional	
max_total_tool_calls	integer	optional	
standard_tool_calls	integer	optional	
critical_tool_calls	integer	optional	
standard_research_model	string	optional	
critical_research_model	string	optional	
max_skills_per_brief	integer	optional	

WebSearchAgent — Research & Discovery

Search the live public web with Auto, Tavily, OpenAI, or Kimi K3. Results are candidates, not verified evidence.

Why it exists: —

Receives: —

Produces: query, requested_provider, actual_provider, fallback_reason, results, result_count, input_tokens, output_tokens

Uses AI: No **Acts outside the platform:** Yes

Field	Type	Req	Description
query	string	required	
provider	string	optional	Live web-search provider.
top_k	integer	optional	
fallback_to_openai	boolean	optional	